

The Iron Rush

I can explain how ironstone, mines, rail, river and ironworks formed a change chain.

[Open the original interactive lesson](#)

Arrival · retrieve the previous move

Standard

Place 1825, 1830 and 1850 in order.

Pupil-chosen arrival variant

Arrival rule: retrieve before adding new content. Personal or family disclosure is never required.

Starter · question before verdict

Enquiry

Why did ironstone found near Eston in 1850 change Teesside so quickly?

Retrieval question: What was found in the Eston Hills in 1850?

- A. A rich seam of ironstone
- B. Ready-made steel beams
- C. Gold

I Do · model the disciplinary move

Model the decision and evidence, not just the answer.

STEP Ironstone is rock containing iron—not finished iron or steel.

STEP It must be mined and moved.

STEP A blast furnace smelts ironstone to make pig iron.

STEP Further ironworking—such as casting, or refining then rolling—made iron products; later additional processing made steel.

Iron chain builder · choose the links in order

I Do · model the disciplinary move

Six links. Start with the rock in the ground. Then test one link with "without it...".

- A. blast furnace
- B. ironstone
- C. ironworking
- D. rail
- E. iron product
- F. pig iron

I Do · model the disciplinary move

Chain: —

A. Without the rail...

B. Without the ironstone...

C. Without the river and port...

Pick a link to test.

We Do · evidence hinge

Commit before reveal: Did the find alone create the boom?

- A. No—mines, workers, transport and ironworks were also needed
- B. Yes—the rock changed itself
- C. No—ironstone had no use

I Do 2 · connect and qualify

Model connection, limitation and honest uncertainty.

STEP Trigger: the 1850 ironstone find.

STEP Conditions: rail, river, existing works and people.

STEP Result: more production and work; exact effects still need evidence.

STEP Material sequence: ironstone was smelted into iron; steel came later through additional processing.

ADAPTED SOURCE CARD

HUM-B-W11-C01 · The 1850 Eston ironstone find

I Do 2 · connect and qualify

Provenance source: HUM-B-W11-S01

Tees Valley Museums records that a rich main seam of ironstone was found in the Eston Hills in 1850. Mines opened, railways carried ironstone to ironworks, and blast furnaces smelted it into iron. Steel production came later.

Can establish Supports the date, place and broad mine–rail–works sequence.

Cannot establish Prove that the discovery alone caused every change or describe every worker's experience.

I Do 2 · connect and qualify

Accessible route Four numbered facts: 1850 find; mines; rail movement; blast furnace makes iron. Ironstone, iron and later steel are different labels.

ADAPTED OFFICIAL-SYNTHESIS CARD

HUM-B-W11-C02 · Industrial growth needs connected conditions

Provenance source: HUM-B-W11-S02

Historic England links Middlesbrough's rapid nineteenth-century growth to coal, the railway, the port, ironstone deposits and iron production. This supports a connected explanation rather than a one-cause story.

I Do 2 · connect and qualify

Can establish Supports broad chronology and the scale of connected industrial change.

Cannot establish Calculate the exact importance of each link or establish household and working conditions.

Accessible route Connection strip: raw material + transport + works + workers → rapid industrial growth.

We Do 2 · interactive humanities lab

Interactive probe: response → evidence or representation → adult receives → next question or scaffold changes.

Iron Chain

CONSTRUCTED TEACHING SOURCE — created for classroom practice. It is not an archival original.

Disciplinary focus: Dependency chain and counterfactual test

Activate the chain, remove one node, then explain the likely bottleneck.

- route Eston Hills ironstone

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- route Mine
- route Rail wagon
- route Blast furnace
- route Pig iron
- route Ironworking: cast, or refine and roll
- route Iron product

RAW

- ironstone

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MOVE

- mine and rail wagon

SMELT

- blast furnace makes pig iron

WORK

- cast, or refine and roll the iron

PRODUCT

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- iron rail
- iron beam

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Evidence classification

- A. RAW ironstone
- B. MOVE mine
- C. MOVE rail wagon
- D. SMELT blast furnace
- E. SMELT pig iron
- F. WORK cast, or refine and roll
- G. PRODUCT iron rail or beam

We Do 2 · interactive humanities lab

Accessible route: Numbered tactile or visual chain: ironstone → mine → rail → blast furnace → pig iron → cast, or refine and roll → iron product, with matching text.

Independent · build defensible evidence

Independent humanities mission

A causal chain separating ironstone, pig iron, finished iron products and later steel, with at least one tested dependency.

16 minutes. Supported, Standard and Stretch are movable routes, not pupil labels.

HUMANITIES STUDIO READY

Source / materials Teacher-prepared constructed lab cards plus the locally approved source cards named for this lesson.

Independent · build defensible evidence

Care / access Use the stated text, visual, pointing, sorting, speaking or exactly scribed route. No personal or family disclosure is required.

Evidence target A causal chain separating ironstone, pig iron, finished iron products and later steel, with at least one tested dependency.

Supported

Order seven chain cards.

Evidence: Chain plus an oral or exactly scribed link.

First _____. Next _____. So _____.

Independent · build defensible evidence

Fade the first prompt no longer needed.

Standard

Add two because-arrows and label the 1850 trigger.

Evidence: Causal chain.

The find mattered because ____ and ____.

Stretch

Remove one link and judge the impact.

Evidence: Counterfactual plus qualification.

Independent · build defensible evidence

Without ____, the rush would be slower because _____. However _____.

Exit · receive, interpret, influence

Standard

Explain two connected links.

VOICE: The pupil explains the ironstone-to-product chain through the selected exit route and chooses POPULATION TILES or STREET PLAN for Week 12.

One BUILD Humanities close only; no signature, receipt, upload queue or second closure form.